

An investigation on the relationship between slit distance and fringe spacing

Jimmy Zeng

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1 Introduction and Background

Wave Phenomena states that wave could interfere when they overlap. When two waves are in opposite phase, it is said to interfere destructively and cancel out. However, if they are the same phase and interferes, it is said to be constructive, reaching a greater amplitude.

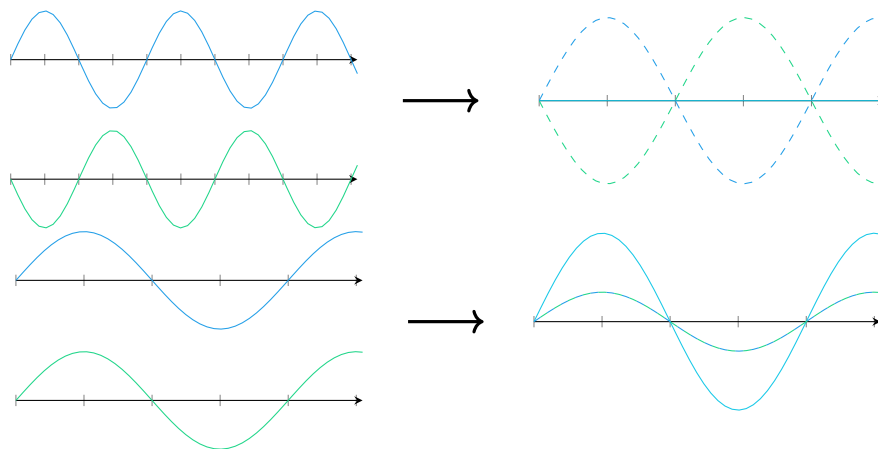


Figure 1: Illustration of wave interference

The debate of whether light is a particle or a wave was challenged by Thomas Young's famous Double Slit Experiment in Young's Course of Lectures on Natural Philosophy and the Mechanical Arts of 1807, where a coherent light source shining through two tiny slits projected an interference pattern with alternation between light and dark fringes rather than two individual bright fringes in a dark background. Illustrated In Figure 2



Figure 2: A Double Slit System

The formula proposed by Thomas Young was

$$s = \frac{\lambda D}{d} \quad (1)$$

In which "S" is the distance between two adjacent bright fringes, λ is the wavelength of the light, D is the distance between the screen and slits in meters and d is the distance between slits. The mathematical geometric derivation is thus justified: Consider a point a on the screen of a double slit system at which two light rays touch from their respective slits. This is illustrated in figure 3

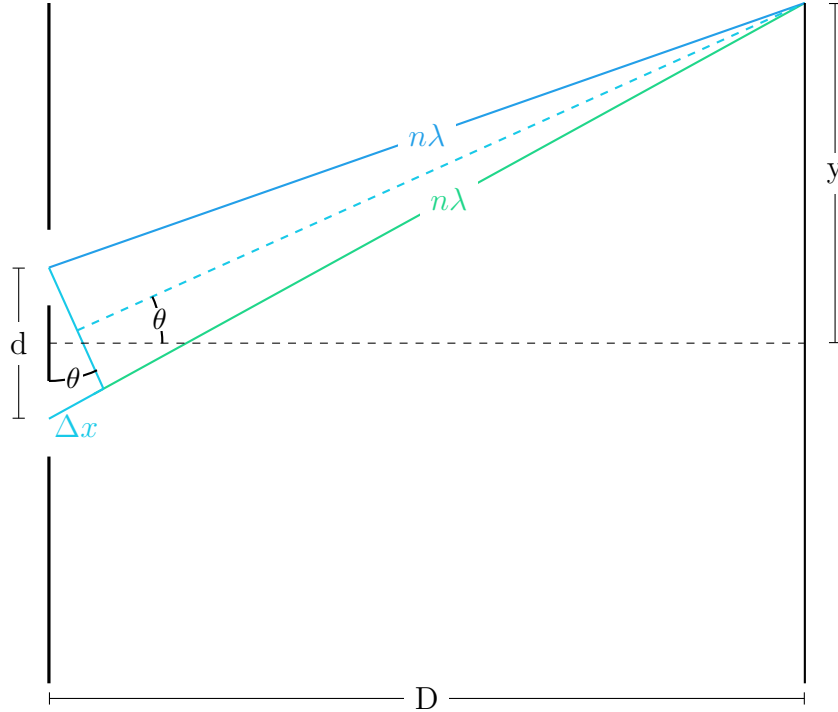


Figure 3: Zoomed in view of double slit system

We observe that the θ that is related by the equation

$$\sin(\theta) = \frac{\Delta x}{d} \quad (2)$$

is equal to the θ represented by

$$\tan(\theta) = \frac{y}{D} \quad (3)$$

When the distance of both light rays' traveled paths is a multiple of the wavelength of the light, they engage in constructive interference. We observe the fact that the path difference (Δx) for constructive waves is

$$n\lambda \tag{4}$$

where λ is the wavelength of the light and n is an integer representing the interested bright fringe. Hence we derive the equation

$$\sin(\theta) = \frac{\Delta x}{d} \tag{5}$$

$$\sin(\theta) = \frac{n\lambda}{d} \tag{6}$$

$$\theta = \sin^{-1} \left(\frac{n\lambda}{d} \right) \tag{7}$$

and since we know that the wavelength λ of light is orders of magnitudes larger than slit spacing d , and hence the value in sin is close to zero, we can use the sin angle approximation $\lim_{\theta \rightarrow 0} \sin(\theta) = \theta$ to simplify to

$$\theta = \frac{n\lambda}{d} \tag{8}$$

relating θ to d and λ . This angle approximation for small values of θ also apply to $\tan(\theta)$ in the other representation of θ

$$\tan(\theta) = \frac{y}{D} \tag{9}$$

and since we know that this is the same θ we can simplify to

$$\theta = \frac{y}{D} \tag{10}$$

and putting them together

$$\frac{y}{D} = \theta \tag{11}$$

$$\frac{n\lambda}{d} = \theta \tag{12}$$

$$\frac{y}{D} = \frac{n\lambda}{d} \tag{13}$$

$$y = n \frac{D\lambda}{d} \tag{14}$$

$$\tag{15}$$

to find the distance between fringes we need to find the distance of a fringe to the central maxima and distance of a consecutive fringe to the central maxima and find the the difference between their distances.

We have defined the distance from the central maximum to

- the n th fringe as $y = n \frac{D\lambda}{d}$
- the n th + 1 fringe as $y = (n + 1) \frac{D\lambda}{d}$

finding the difference $\Delta y = s$ gives us the spacing between adjacent fringes

$$\Delta y = s = (n + 1) \frac{D\lambda}{d} - n \frac{D\lambda}{d} \quad (16)$$

$$s = (n + 1 - n) \frac{D\lambda}{d} \quad (17)$$

$$s = \frac{D\lambda}{d} \quad (18)$$

and now we arrive at our destination.

2 Methodology

2.1 Research Hypothesis and Method variables

The independent variable is the distance D between the slit and the screen. There will be two experiments carried out with different setups, namely, the "macro" and "micro" experiments. This is to satisfy the trial quantity to increase validity and each experimental procedure explained in the ensuing section.

- Micro will solely focus on distances within a meter and will include (in cm) 20, 40, 60, 80, and 100.
- Macro will be done from various distances ranging from 3 to 8 meters and will include (in m) 3, 5, 6, and 8.
 - Each of these distances will have micro steps that include (in meters) +0, +50, and +100 which will extend our trial amounts
 - For example, for 5 meters we will have (in m) 5.00, 5.50, and 6.00.

What is the relationship with the distance between screen and slit (from the range of 20cm to 8 m) and the spacing between adjacent bright fringes (in mm) in a double slit system.

The measured dependent variable is the distance s between two adjacent bright fringes. Illustrated by the following diagram. And hence our guiding question is

And we hypothesize that the relationship would be a linear relationship with the distance between screen and slit and the spacing between adjacent bright fringes. This is because our mathematically derived equation $s = D \frac{\lambda}{d}$ suggests a directly proportional relationship between s and D .

2.2 Variable Control and Measurement

The distance between slit and screen will be modified two means.

The long range consists of the aforementioned 3m, 5m, 6m, and 8m will be achieved with the placement of three tables (of which the recording instrument will hang on) in those respective distances, the last 8m is the distance from the origin directly to the wall. They will be relative to an arbitrary "origin" point in the classroom marked with a masking tape.

The short range will represent the aforementioned "micro steps" (0cm, 50cm, and 100cm) and will be achieved by placing the slit (and light source) on a momentum test cart (this is because they move back and forth very smoothly on their designated tracks).

2.3 Procedure Consideration and Setup Description

Since the fringe spacing is within the millimeter range and it is beyond my human ability to hold the clipboard still within millimeter stillness while using the other hand to hold a pencil and mark accurately on the paper live. I decided to take a high resolution photo of the projection on a graphed piece of paper as a photo is most versatile and captures exactly as seen. The graphed piece of paper is printed with millimeter accuracy markings. As a result I will measure distance between each adjacent fringe (in millimeters) with software solutions and excuse my skill issue.